

3.8. Box-Whisker Plot

A box plot or box-whisker plot is a set of five summary measures of the set of data; they are (i) median (ii) lower quartile (iii) upper quartile (iv) smallest observation (v) largest observation.

What makes the box plot (also called box-whisker-plot) a very useful graph is that it shows all of the above five numbers. Thus, it only takes some seconds to reveal the central tendencies and dispersion of one's data. Additionally, a box plot may also indicate any outliers.

For a given set of data, construction of box plot starts with the calculation of first, second and third quartiles Q_1 , Q_2 , and Q_3 of the data. Quartiles are the values which divide the entire data into four equal parts. Consider the data set contains N observations. To identify Q_1 , Q_2 , and Q_3 , first we arrange the data in ascending order of magnitude, then locate the value which is coming in the $\left(\frac{N}{4}\right)^{th}$,

$\left(\frac{N}{2}\right)^{th}$ and $\left(\frac{3N}{4}\right)^{th}$ position. These values are respectively Q_1 , Q_2 , and Q_3 .

Let us consider the following data showing the duration rounded in hours of 45 hospital patients slept following the administration of a certain anesthetic

6,10,12,4,8,7,6,8,5,12,11,3,8,7,2,13,8,4,4,5,5,8,7,7,3,2,3,8,9,6,7,10,3,6,5,5,3,4,17,9,4,7,7,11 and 8.

Arranging the data in ascending order of magnitude, it is as follows:

2,2,3,3,3,3,6,4,4,4,9,4,5,5,5,5,6,6,6,6,7,7,7,7,7,7,7,8,8,8,8,8,8,8,9,9,10,
10,11,11,12,12,13,17

That is expressed in an ungrouped frequency table as,

Duration rounded in hours	Number of patients	Cumulative frequency
2	2	2
3	4 ⁵	6 ⁷
4	4 ⁵	10 ¹²
5	5	15 ¹⁷
6	5 ⁴	20 ²¹
7	7	27 ²⁸
8	7	34 ³⁵
9	3 ²	37 ³⁷
10	2	39 ³⁹
11	2	41
12	2	43
13	1	44
17	1	45

Here $\frac{N}{4} = \frac{45}{4} = 11.25$. From the data, the cumulative frequency

gives, the observations from 10th to 15th are 5s.

Hence $Q_1 = \left(\frac{N+1}{4}\right)^{th}$ value is 4.

Here $\frac{N+1}{2} = \frac{45}{2} = 22.5$. From the data, the cumulative

frequency gives, the observations from 20th to 27th are 7s.

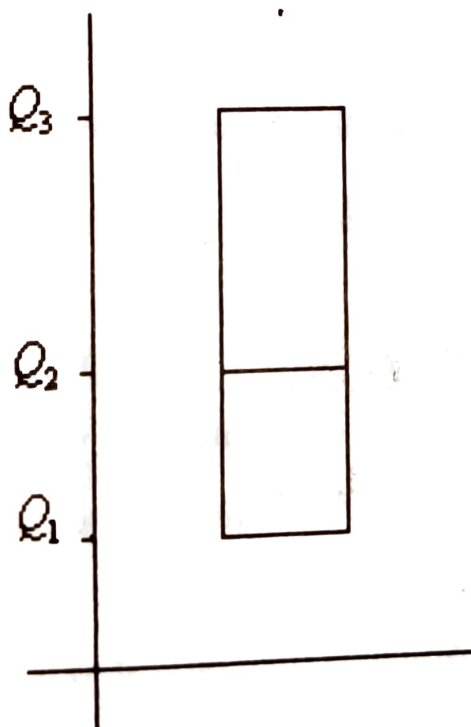
Hence $Q_2 = \left(\frac{N+1}{2}\right)^{th}$ value is 7.

Here $\frac{3N}{4} = \frac{3 \times 45}{4} = 33.75$. From the data, the cumulative

frequency gives, the observations from 27th to 34th are 8s.

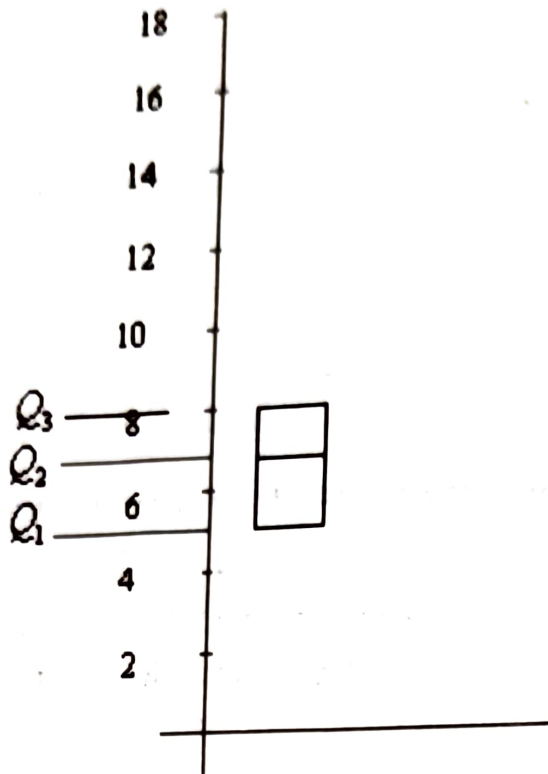
Hence $Q_3 = \left(\frac{3N}{4} \right)^{th}$ value is 8.

To draw box plot for a set of data, a box is drawn in the graph by considering Q_1 as the bottom and Q_3 as the top of the box. A horizontal line which partition the box is also drawn through the value of Q_2 . For a data, the box is as seen below:



If horizontal line partitioning the box corresponding to Q_2 divides the box in equal parts, the observations are considered symmetric.

For the given data, the box is as follows:



Now we identify some more values: they are

- (i) H-spread; which is $Q_3 - Q_1$
- (ii) 1.5 X H-spread
- (iii) $Q_3 + 1.5 \times H$ - spread, which is called Upper Inner Fence
- (iv) $Q_1 - 1.5 \times H$ - spread, which is called Lower Inner Fence
- (v) $Q_3 + 3 \times H$ - spread, which is called Upper Outer Fence
- (vi) $Q_1 - 3 \times H$ - spread, which is called Lower Outer Fence
- (vii) Upper Adjacent, which is the largest observation below the upper inner fence.
- (viii) Lower Adjacent, which is the smallest observation above the lower inner fence.
- (ix) Outside values, which are the values beyond an inner fence but not beyond outer fence and
- (x) Far outside values or extreme values, which are the values beyond an outer fence.

Outside values are called outliers. They are considered as unusual values in a set of observations.

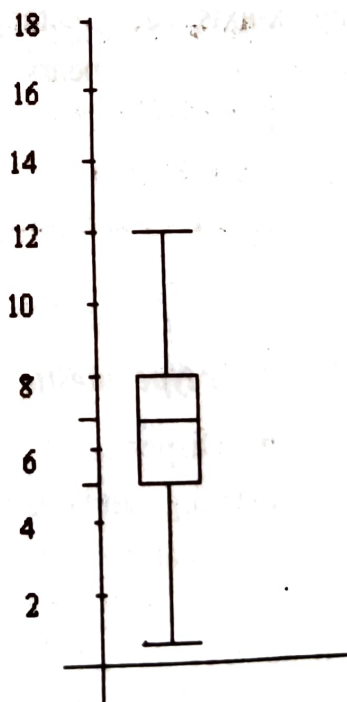
For the above considered set of observations,

- (i) $H\text{-spread} = 8 - 5 = 3.$
- (ii) $1.5X\ H\text{-spread} = 4.5$
- (iii) $\text{Upper Inner fence} = 8 + 4.5 = 12.5$
- (iv) $\text{Lower Inner Fence} = 5 - 4.5 = 0.5$
- (v) $\text{Upper Outer Fence} = 8 + 9 = 17$
- (vi) $\text{Lower Outer Fence} = 5 - 9 = -4$ (Which is not possible value of the data set)
- (vii) $\text{Upper Adjacent} = \text{largest observation less than } 12.5 = 12$
- (viii) $\text{Lower Adjacent} = \text{least observation above } 0.5 = 2$

Outside values (Outliers) = Values within $12.5 - 17$; and values less than 0.5 . They are 13 and 17 . No far outside or extreme values are observed.

To complete the box plot we draw lines perpendicular to lower and upper end of the rectangles, which are called whiskers, in such a way that the perpendicular drawn from upper end of the rectangle is extended up to the upper adjacent and the perpendicular drawn from the lower end of the rectangle is extended up to the lower adjacent.

In a box plot every observation are not marked, but outliers are represented by circles. They are unusual values lying outside the range where approximately 99% of the observations lie.

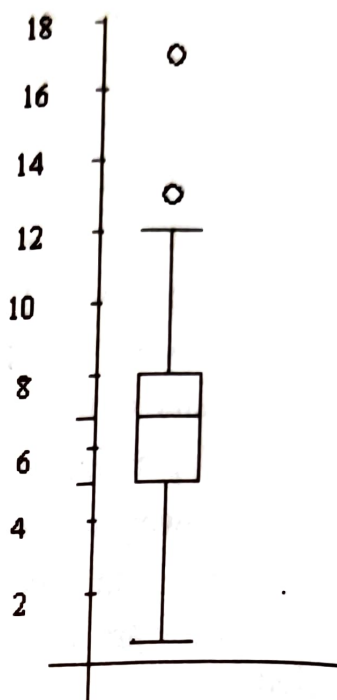


Since the adjacent of the data considered are 12 and 0.5 , the whiskers are drawn as follows:

The outliers are also marked as circles to complete the box-whisker plot of the given data. The plot is as follows:

Examination of a box and whisker plot for a set of data reveals information regarding the amount of spread, location of concentration and symmetry of the data. More than two related sets of data can be represented in same graph in parallel by box-whisker plots for a better comparison.

If the data is considered along x-axis, the rectangle in box-whisker plot appears parallel to x-axis.



Box-whisker plot showing the duration rounded in hours of 45 hospital patients slept following the administration of a certain anesthetic.

EXERCISES

Short answer type questions

1. Define Dispersion.
2. Define range and quartile deviation.
3. Define Mean deviation about median.
4. What is meant by R.M.S.D?
5. If the S.D. of set of observations 5, 8, 10, 14, 17 is k , what is the S.D. of set of observations 7, 10, 12, 16, 19? Ans: k
6. The variance of a set of n consecutive numbers is 14. Find n .
7. Define Outliers.
8. Define Upper Outer Fence and Lower Outer Fence.
9. Define H-spread.